

Python Cheat Sheet

Base Types	
<i>integer, float, boolean, string, bytes</i>	
int	163 0 -192 $\underbrace{0b110}_{\text{binary}}$ $\underbrace{0x3F}_{\text{hex}}$
float	9.32 0.0 $-1.7E-6$ $\times 10^{-6}$
bool	$\underbrace{\text{False}}_0$ $\underbrace{\text{True}}_1$
str	'some text' or "some text"
bytes	b"text\xfe\775"

Container Types	
ordered containers — repeatable values	
list	[1,5,3] ["a",1,5,5] [5] []
tuple	(1,5,3) "a",1,5,5 (5,) ()
Immutable (non-modifiable values)	
str	"153" ""
key containers — no order, unique keys	
set	{"key1", "key2"} {1,9,3,0} set()
dict	{"key1":value1, "key2":value2} dict(a=3,b="v") {}

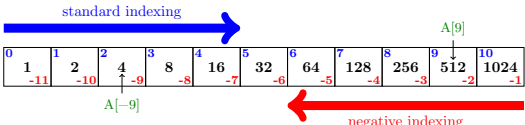
Integer Sequences
range ([start,] end [,step])
start default is 0 (inclusive), end (exclusive), step default is 1.
range (5) → 0,1,2,3,4
range (2,5) → 2,3,4
range (2,12,3) → 2,5,8,11
range (20,5,-5) → 20,15,10

Operations on Sets
Operators
.union
& .intersection
- .difference
Methods
s.add(key) s.update(s2)
s.clear() s.remove(key)

Operations on Lists
Methods
a.append(value) a.extend(a2)
s.insert(idx,value) a.pop()

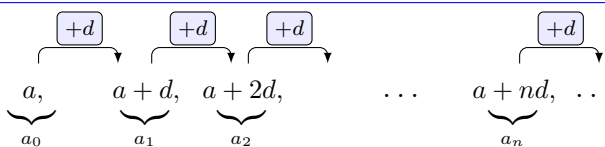
Conversions
type (expression)
int ('153') → 15
int ('3f',16) → 63 (Specify base in 2 nd parameter)
int (-11.24e8) → -1124000000
int (15.56) → 15 (Truncate decimal point)
round (15.58,1) → 15.6 (Round to 1 decimal place)
float ('15.56') → 15.56
bool (x) (False for None, zero or empty containers)
str (x) (String representation of x.)
chr (65) → 'A' code ↔ char ord ('A') → 65
list ('abc') → ['a','b','c']
dict ([(3,'three'), (1,'one')]) → {3:'three', 1:'one'}
set (['one','two']) → {'one','two'}
(Split string using a separator, str → list of str)
'random:data:666'.split(':') → ['random', 'data', '666']
(Join a list of strings, list of str → str)
':'.join(['random', 'data', '666']) → 'random:data:666'
(Convert each element in a collection)
[int (x) for x in ['1','29','-3']] → [1, 29, -3]

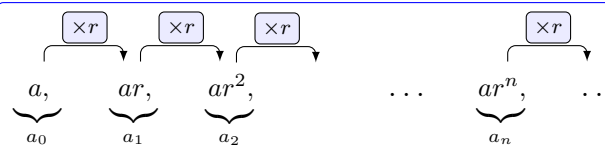
Generic Operations on Containers
min (c) max (c) sum (c) sorted (c)
len (c) (Number of elements in collection c)
all (c) → True if all items in c evaluate to True, else False.
any (c) → True if at least one item in c evaluate to True, else False.

Sequence Containers Indexing
standard indexing

A [9]
A [-9]
negative indexing
a[3:6] → [8, 16, 32]
a[1:-1] → [2, 4, 8, 16, 32, 64, 128, 256, 512]
a[::-1] → [1024, 512, 256, 128, 64, 32, 16, 8, 4, 2, 1]

Looping over Collections		
(Loop over values)	(Count and loop over values)	(While loop)
for value in A:	for k,value in enumerate (A):	k = 0
print (value)	print (k, value)	while k< len (A):
		print (k, A[k])
		k += 1
Initialisation before loop.		
update within loop.		
break immediatly exits loop. continue skips to next iteration.		
else block for normal loop exit.		

AP and GP

AP

Recursive definition: $a_n = a_{n-1} + d$ with $a_0 = a$.
Closed formula: $a_n = a + nd$.

GP

Recursive definition: $a_n = ra_{n-1}$ with $a_0 = a$.
Closed formula: $a_n = ar^n$.